

Introduction to Energy

Name: _____

Date: _____

A. Kinematics Model Review

1) Kinematics describes motion using four key quantities. What are they?

_____, _____, _____, _____

2) What are the units for each of the four key quantities?

_____, _____, _____, _____

3) The two most common graphs used in kinematics are:

a) _____

b) _____

4) A student says: “If the slope of a position–time graph is positive, the object must be speeding up.” Is that correct? Explain in one sentence.

B. Forces Model Review

5) In what way are forces represented graphically? _____

6) How do we find the **net force** on an object? _____

C. Why We Need a New Model

Consider the following situations:

- a ball rolling down a hill
- a pendulum swinging
- a cart compressing a spring

7) In each case, the *forces and directions* change as the motion occurs. What might be easier to track than “every force at every moment”?

8) What do you think is being *transferred* or *transformed* in all three examples?

D. Defining Energy

Energy is the ability to **do work**.

Mechanical Energy is energy associated with **motion** or **potential motion** of an object.

9) What are some real-world examples of things that require energy or do work? (List five.)

- _____
- _____
- _____
- _____
- _____

Types of Mechanical Energy (Kinetic & Potential)

1) **Kinetic Energy K or KE** is energy due to **motion**.

Any object that has a **velocity** has kinetic energy.

Example:

2) **Gravitational Potential Energy U_g or PE** depends on an object's vertical displacement, also called **height**.

Example:

- 3) **Elastic Potential Energy** U_s or PE_s depends on how much something is **stretched** or **compressed**.

Example:

There are other forms of energy as well:

- 1) **Thermal Energy** (heat) is internal energy of a substance due to the movement of its atoms and molecules.
- 2) **Chemical Energy** is energy stored in the bonds of atoms and molecules, released during chemical reactions (e.g., fuel, food).
- 3) **Nuclear Energy** is energy stored in the nucleus of an atom, released by splitting (fission) or combining (fusion) atoms.

E. Practice: Identifying Mechanical Energy

Directions: For each situation, identify the type(s) of mechanical energy present **at that instant**. Circle all that apply.

- | | | | |
|---|---|----|----|
| 1) A book resting on a high shelf: | K | Ug | Us |
| 2) A skateboard moving on level ground: | K | Ug | Us |
| 3) A stretched rubber band held at rest: | K | Ug | Us |
| 4) A ball falling straight down (mid-air): | K | Ug | Us |
| 5) A cart momentarily at rest at the top of a hill: | K | Ug | Us |
| 6) A compressed spring on a table: | K | Ug | Us |

F. Energy Pie Charts (Model Introduction)

In kinematics, motion is modeled using **position–time** and **velocity–time** graphs.

In forces, interactions are modeled using **free-body diagrams**.

In energy, we model situations using *energy pie charts*.

Energy pie charts:

- Represent the **total mechanical energy** of a system. (Energy within a closed system is conserved. Energy within an open system is lost).
- Show how energy is **distributed among different types**
- Do **not** show direction or speed
- Are used at **specific instants in time**

G. How to Build an Energy Pie Chart

Follow these steps each time:

1. Define the system
 - A system is any object or place energy can be stored or used
2. Define reference point(s):
 - U_g depends upon an object's height. Where is $h = 0\text{m}$?
 - U_s depends upon stretch or compression. Where is the spring at its resting position?
3. Determine types of energy present at specific instances. Decide whether each energy is:
 - zero
 - some
 - most or all
4. Make sure the pie represents the **energy at that instant only**.
5. Conservation:
 - Is total energy conserved?
 - Is mechanical energy conserved?

Example

A ball is dropped from rest and is halfway to the ground.

- | | |
|-----------------------------|----------|
| • Does it have velocity? | Yes / No |
| • Does it have height? | Yes / No |
| • Does it involve a spring? | Yes / No |

Energy present (circle):	K	Ug	Us
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How does the pie chart look like?

H. Practice: Multiple Energy Types

Directions: Some situations involve more than one type of mechanical energy.

- | | | | |
|---|---|----|----|
| 1) A ball rolling down a hill: | K | Ug | Us |
| 2) A pendulum at its lowest point: | K | Ug | Us |
| 3) A skier halfway down a slope: | K | Ug | Us |
| 4) A spring launcher while the cart is moving: | K | Ug | Us |
| 5) A ball thrown straight upward (on the way up): | K | Ug | Us |
| 6) A diver just before hitting the water: | K | Ug | Us |

I. Practice: Energy Pie Charts (Qualitative)

Directions: For each scenario below, the motion is shown at several **specific instants in time**.

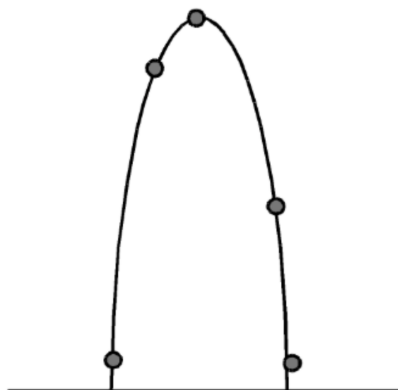
For *each instant*:

- Identify which types of mechanical energy are present
- Draw an **energy pie chart** showing the relative amounts of:

$$K \quad U_g \quad U_s$$

- Assume mechanical energy is conserved unless otherwise stated

1. Projectile Motion



Instants shown:

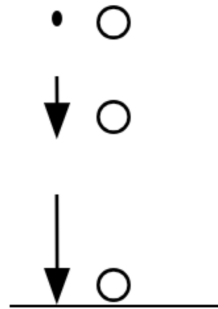
- Just after launch
- At the highest point
- Just before landing

Energy types present at each instant (circle all that apply):

- | | | | |
|-------------------|---|-------|-------|
| • Launch: | K | U_g | U_s |
| • Highest point: | K | U_g | U_s |
| • Before landing: | K | U_g | U_s |

Draw energy pie charts:

2. Object in Free Fall



Instants shown:

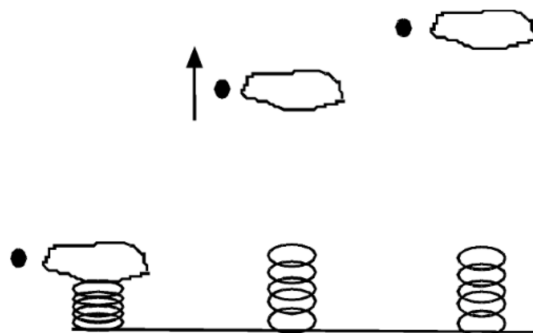
- At release
- Midway down
- Just before hitting the ground

Energy types present at each instant (circle all that apply):

- | | | | |
|------------------|---|----------------|----------------|
| • Release: | K | U _g | U _s |
| • Midway: | K | U _g | U _s |
| • Before impact: | K | U _g | U _s |

Draw energy pie charts:

3. Spring Launcher



Instants shown:

- Spring fully compressed
- Spring partially compressed
- Spring relaxed, object moving

Energy types present at each instant (circle all that apply):

- | | | | |
|-------------------------|---|----------------|----------------|
| • Fully compressed: | K | U _g | U _s |
| • Partially compressed: | K | U _g | U _s |
| • Moving freely: | K | U _g | U _s |

Draw energy pie charts:

4. Object Moving Along Level Ground



Instants shown:

- Initial motion
- Later motion

Energy types present at each instant (circle all that apply):

- | | | | |
|------------|---|----------------|----------------|
| • Initial: | K | U _g | U _s |
| • Later: | K | U _g | U _s |

Draw energy pie charts:

Why does the gravitational potential energy not change in this situation?